

LLM-Identity Decomposition: Per-Model Asymmetry within HA and AA

Does dyad asymmetry depend on which LLM is in the conversation?

PENPAL Project

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1 Overview

The cross-condition notebooks treat HA and AA as monoliths. This notebook decomposes both by which LLM is in the dyad and asks two questions per metric:

1. **Per-LLM magnitude.** Does unsigned asymmetry A_m vary across the four LLMs within HA? Within AA?
2. **Per-LLM direction.** Does signed slot-asymmetry vary across LLMs (i.e. are some models more likely to drive in a particular direction)?

Sample sizes per LLM are small (HA: ~22-29 stories per model; AA: 20 per model), so reported tests are non-parametric (Kruskal-Wallis + pairwise Wilcoxon BH-adjusted). Bootstrap CIs are also reported.

Where it makes sense, the same-model rows in HA and AA are placed side-by-side: does Claude paired with a human exhibit the same asymmetry profile as Claude paired with itself?

2 Build per-metric per-story tables

```
## Per-story tables built:
##   valence: 180 (HA=100, AA=80)
```

novelty: 180 (HA=100, AA=80)
semexp: 177 (HA=97, AA=80)

3 Within-HA: per-LLM Asymmetry

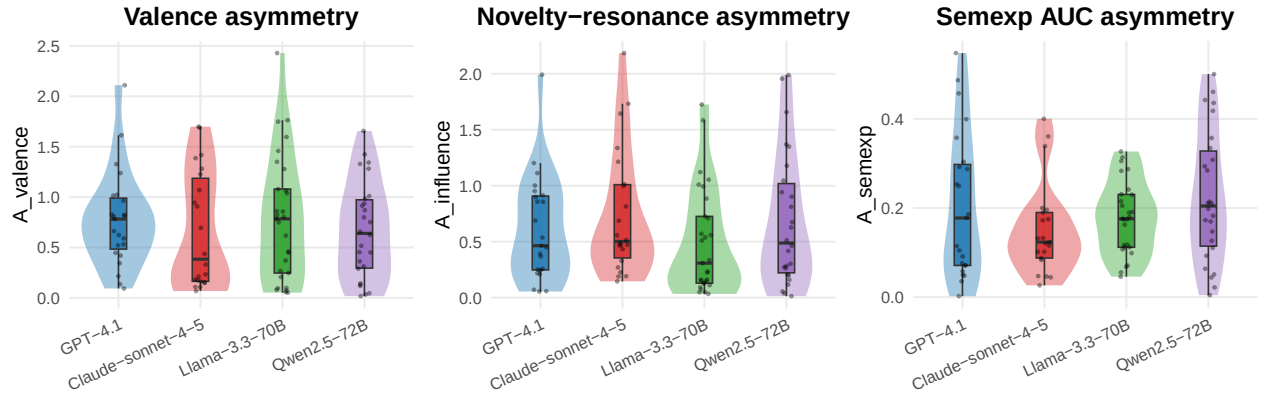
3.1 Magnitude

Table 1: Per-LLM unsigned asymmetry magnitude within HA

metric	llm	n	mean	median	sd
A_valence	GPT-4.1	23	0.780	0.782	0.475
A_valence	Claude-sonnet-4-5	22	0.665	0.384	0.577
A_valence	Llama-3.3-70B	29	0.785	0.784	0.622
A_valence	Qwen2.5-72B	26	0.671	0.640	0.481
A_influence	GPT-4.1	23	0.611	0.464	0.470
A_influence	Claude-sonnet-4-5	22	0.744	0.502	0.564
A_influence	Llama-3.3-70B	29	0.496	0.308	0.471
A_influence	Qwen2.5-72B	26	0.685	0.488	0.593
A_semexp	GPT-4.1	23	0.205	0.178	0.162
A_semexp	Claude-sonnet-4-5	21	0.152	0.123	0.103
A_semexp	Llama-3.3-70B	27	0.177	0.176	0.082
A_semexp	Qwen2.5-72B	26	0.225	0.204	0.147

Table 2: Within-HA Kruskal-Wallis: $A_m \sim \text{llm}$

metric	kw_p
A_valence	0.8160
A_influence	0.2696
A_semexp	0.3815



3.2 Direction

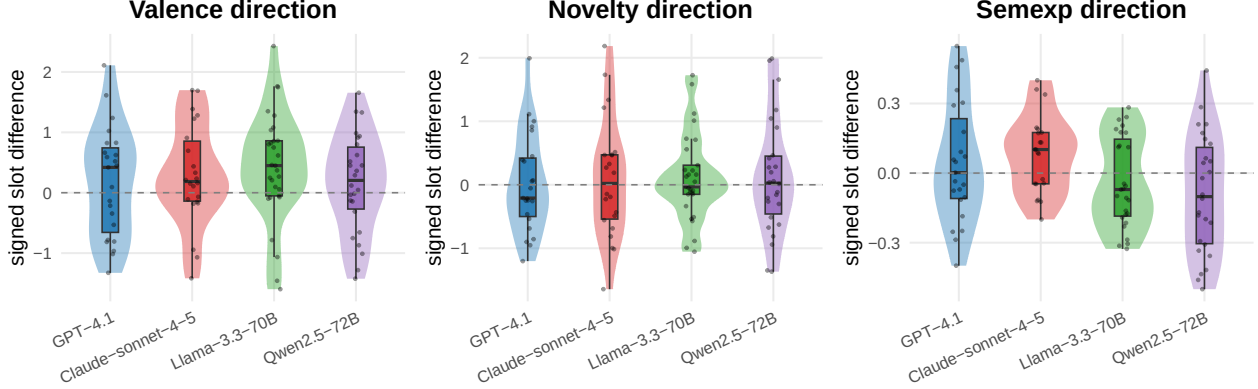
Per-LLM directional stability of the SIGNED slot difference within HA. If the human-side identity is what drives directionality, all four LLMs should look similar (same signed direction). If specific LLMs shape directionality (e.g., Claude is more deferential than Qwen), the stability index will differ across models.

Table 3: Within-HA directional stability of signed asymmetry per LLM

metric	llm	n	mean	sd	stability	sign_consistency	one_sample_t_p
valence	GPT-4.1	23	0.176	0.910	0.194	0.565	0.363
valence	Claude-sonnet-4-5	22	0.291	0.841	0.346	0.636	0.119
valence	Llama-3.3-70B	29	0.425	0.915	0.465	0.724	0.018
valence	Qwen2.5-72B	26	0.161	0.821	0.196	0.577	0.326
novelty	GPT-4.1	23	0.017	0.781	0.021	0.565	0.920
novelty	Claude-sonnet-4-5	22	0.090	0.944	0.095	0.500	0.660
novelty	Llama-3.3-70B	29	0.104	0.682	0.153	0.517	0.417
novelty	Qwen2.5-72B	26	0.121	0.908	0.133	0.538	0.505
semexp	GPT-4.1	23	0.046	0.261	0.178	0.522	0.402
semexp	Claude-sonnet-4-5	21	0.083	0.166	0.500	0.619	0.033
semexp	Llama-3.3-70B	27	-0.041	0.194	0.211	0.630	0.283
semexp	Qwen2.5-72B	26	-0.090	0.257	0.351	0.577	0.086

Table 4: Variance-equality across LLMs (within HA)

metric	brown_forsythe_p
valence	0.8809
novelty	0.3798
semexp	0.0903



4 Within-AA: per-Model Asymmetry

The AA dataset only contains same-model self-pairings (no cross-model pairs). The four levels are therefore pure within-model ai-vs-ai conversations.

4.1 Magnitude

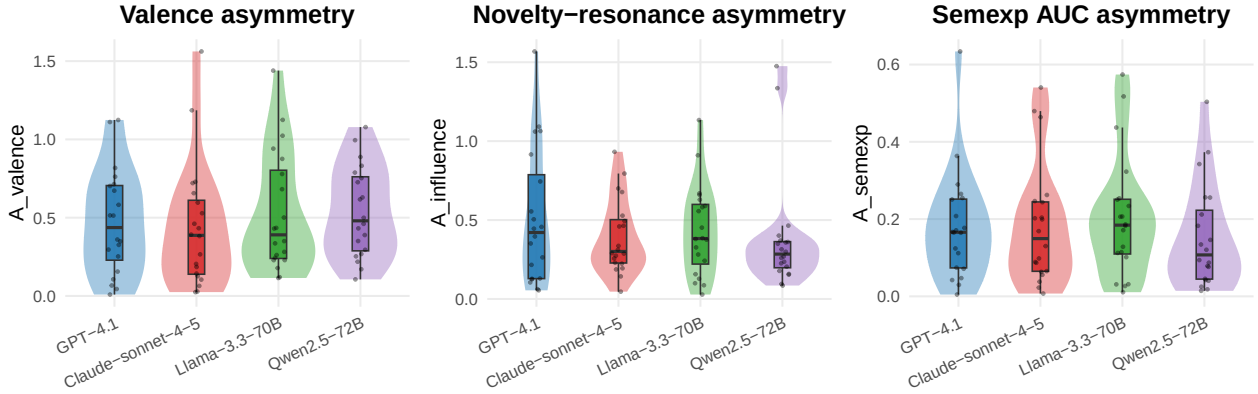
Table 5: Per-model unsigned asymmetry magnitude within AA

metric	llm	n	mean	median	sd
A_valence	GPT-4.1	20	0.474	0.437	0.335
A_valence	Claude-sonnet-4-5	20	0.440	0.387	0.397

metric	llm	n	mean	median	sd
A_valence	Llama-3.3-70B	20	0.529	0.390	0.380
A_valence	Qwen2.5-72B	20	0.537	0.480	0.288
A_influence	GPT-4.1	20	0.512	0.421	0.429
A_influence	Claude-sonnet-4-5	20	0.392	0.301	0.235
A_influence	Llama-3.3-70B	20	0.434	0.383	0.287
A_influence	Qwen2.5-72B	20	0.380	0.284	0.366
A_semexp	GPT-4.1	20	0.181	0.166	0.144
A_semexp	Claude-sonnet-4-5	20	0.183	0.149	0.156
A_semexp	Llama-3.3-70B	20	0.205	0.184	0.158
A_semexp	Qwen2.5-72B	20	0.152	0.107	0.135

Table 6: Within-AA Kruskal-Wallis: $A_m \sim \text{model}$

metric	kw_p
A_valence	0.5618
A_influence	0.6659
A_semexp	0.7215



4.2 Direction

Table 7: Within-AA directional stability of signed asymmetry per model

metric	llm	n	mean	sd	stability	sign_consistency	one_sample_t_p
valence	GPT-4.1	20	-0.117	0.578	0.203	0.65	0.376
valence	Claude-sonnet-4-5	20	-0.106	0.591	0.180	0.60	0.431
valence	Llama-3.3-70B	20	0.134	0.648	0.207	0.65	0.366
valence	Qwen2.5-72B	20	0.176	0.596	0.295	0.60	0.203
novelty	GPT-4.1	20	0.052	0.676	0.078	0.55	0.733
novelty	Claude-sonnet-4-5	20	0.081	0.458	0.177	0.55	0.438
novelty	Llama-3.3-70B	20	-0.165	0.502	0.328	0.70	0.159
novelty	Qwen2.5-72B	20	-0.229	0.480	0.477	0.70	0.046
semexp	GPT-4.1	20	0.014	0.234	0.061	0.50	0.788
semexp	Claude-sonnet-4-5	20	0.006	0.244	0.026	0.50	0.910
semexp	Llama-3.3-70B	20	-0.020	0.262	0.078	0.60	0.733
semexp	Qwen2.5-72B	20	0.042	0.202	0.207	0.60	0.365

Table 8: Variance-equality across models (within AA)

metric	brown_forsythe_p
valence	0.9052
novelty	0.4035
semexp	0.7469

5 Same Model Across Conditions: HA vs AA per-LLM

For each LLM, place its HA stories (paired with a human) next to its AA stories (paired with itself). Asks whether a given model’s asymmetry profile depends on partner type.

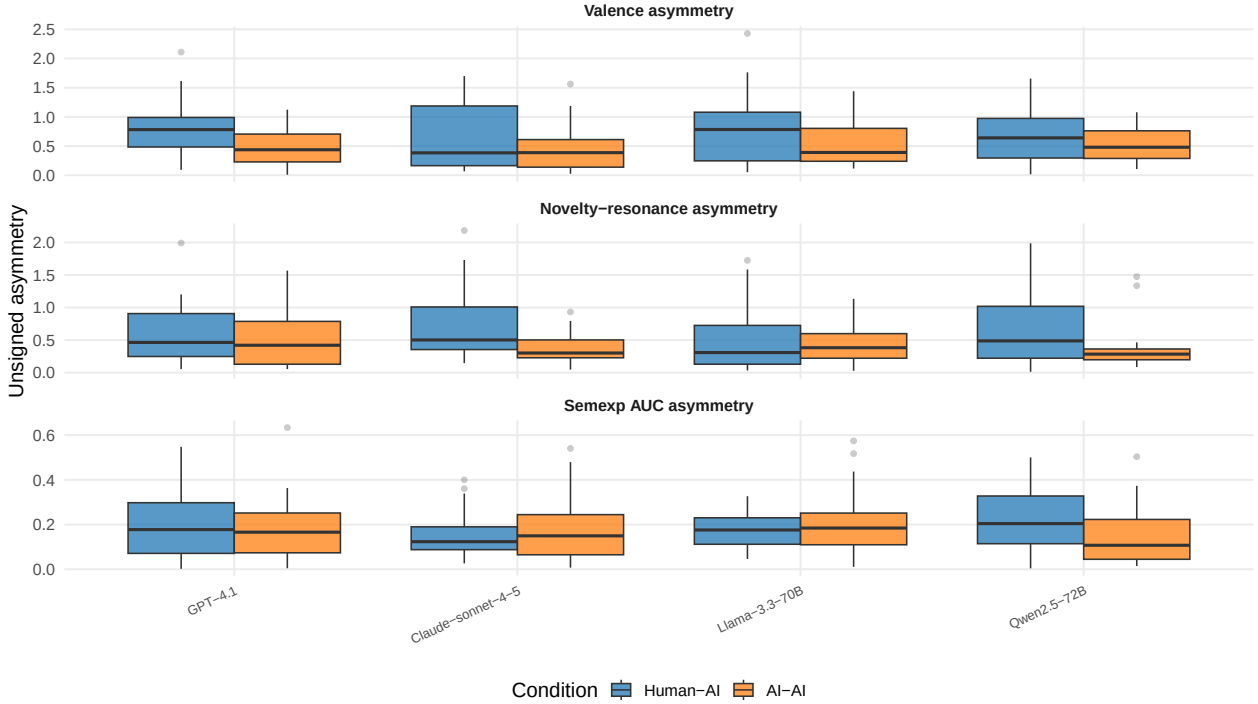


Table 9: HA vs AA per-LLM Wilcoxon (BH-adjusted within metric)

metric	llm	n_HA	n_AA	median_HA	median_AA	cliff_delta	wilcox_p	p_BH
valence	GPT-4.1	23	20	0.782	0.437	0.409	0.022	0.087
valence	Claude-sonnet-4-5	22	20	0.384	0.387	0.200	0.276	0.372
valence	Llama-3.3-70B	29	20	0.784	0.390	0.186	0.279	0.372
valence	Qwen2.5-72B	26	20	0.640	0.480	0.135	0.448	0.448
novelty	GPT-4.1	23	20	0.464	0.421	0.109	0.555	0.739
novelty	Claude-sonnet-4-5	22	20	0.502	0.301	0.391	0.030	0.121
novelty	Llama-3.3-70B	29	20	0.308	0.383	-0.038	0.832	0.832
novelty	Qwen2.5-72B	26	20	0.488	0.284	0.277	0.114	0.227
semexp	GPT-4.1	23	20	0.178	0.166	0.083	0.656	0.840
semexp	Claude-sonnet-4-5	21	20	0.123	0.149	-0.067	0.728	0.840
semexp	Llama-3.3-70B	27	20	0.176	0.184	-0.037	0.840	0.840
semexp	Qwen2.5-72B	26	20	0.204	0.107	0.304	0.082	0.328

6 Discussion-friendly summary tables

Quick-read summary across all metrics: which LLMs (if any) stand out on magnitude, on direction, and on partner-sensitivity?

Table 10: Directional stability per LLM, by condition (Human-AI vs AI-AI)

condition	metric	llm	n	stability	sign_consistency	one_sample_t_p
AI-AI	novelty	GPT-4.1	20	0.078	0.550	0.733
AI-AI	novelty	Claude-sonnet-4-5	20	0.177	0.550	0.438
AI-AI	novelty	Llama-3.3-70B	20	0.328	0.700	0.159
AI-AI	novelty	Qwen2.5-72B	20	0.477	0.700	0.046
AI-AI	semexp	GPT-4.1	20	0.061	0.500	0.788
AI-AI	semexp	Claude-sonnet-4-5	20	0.026	0.500	0.910
AI-AI	semexp	Llama-3.3-70B	20	0.078	0.600	0.733
AI-AI	semexp	Qwen2.5-72B	20	0.207	0.600	0.365
AI-AI	valence	GPT-4.1	20	0.203	0.650	0.376
AI-AI	valence	Claude-sonnet-4-5	20	0.180	0.600	0.431
AI-AI	valence	Llama-3.3-70B	20	0.207	0.650	0.366
AI-AI	valence	Qwen2.5-72B	20	0.295	0.600	0.203
Human-AI	novelty	GPT-4.1	23	0.021	0.565	0.920
Human-AI	novelty	Claude-sonnet-4-5	22	0.095	0.500	0.660
Human-AI	novelty	Llama-3.3-70B	29	0.153	0.517	0.417
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Human-AI	valence	Llama-3.3-70B	29	0.465	0.724	0.018
Human-AI	valence	Qwen2.5-72B	26	0.196	0.577	0.326

7 Summary

What this notebook adds:

- **Within-HA per-LLM:** tests whether the four LLMs differ in unsigned asymmetry A_m (KW + per-LLM bootstrap-style CIs in the magnitude table) and in directional stability of the signed slot difference. If the HA-specific identity-tied directionality reading from the main notebook holds, all four LLMs should show high stability and a consistent sign on each metric. If it breaks down for one LLM, that LLM contributes less to the HA-distinct signature.
- **Within-AA per-model:** tests whether self-pairings of different LLMs produce different asymmetry profiles. A pure mathematical-baseline reading predicts no difference; a model-specific stylistic reading predicts variation (e.g., Claude self-pair more aligning, Qwen self-pair more divergent).
- **HA vs AA per-LLM:** holds the LLM constant and varies the partner. Same-model HA-vs-AA differences (Wilcoxon + Cliff’s δ , BH-adjusted per metric) isolate the effect of swapping a human partner for a same-LLM partner.

Article-use notes: - Use the Within-HA per-LLM table to support a sentence in the Discussion on whether the HA finding is model-general or model-driven. - Use Within-AA per-LLM to qualify the structural-null

result: if AA's rubber-band excess is concentrated in one model (likely Claude or GPT), this reads as model-specific homogenization, not "AI-AI dyads in general". - Use HA vs AA per-LLM as a partner-sensitivity panel that complements the three primary cross-condition tests.

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